

AMENDMENT NO. 1 NOVEMBER 2020
TO
IS 8472 : 2019 CENTRIFUGAL REGENERATIVE PUMP FOR CLEAR
COLD WATER — SPECIFICATION
(Second Revision)

(Page 6, clause **12.3.1**, line 2 and 6) — Substitute ‘12.3’ for ‘13.3’.

(Page 6, clause **12.5**, line 2 and 3) — Delete ‘a minimum static lift of 1.5 m for semi self-priming and.

(Page 6, clause **12.5**, line 3) — Substitute ‘4m’ in place of ‘3m’.

(Page 7, clause **13.1.2**, line 3) — Substitute ‘13.1.2.1’ for ‘14.1.2.1’.

(Page 7, clause **13.1.2.1.1**, line 5) — Substitute ‘IS 15999 Part 1 / IEC 60034-1’ for ‘IS/IEC 60034-1’.

(Page 7, clause **13.1.2.1.2**, line 3) — Substitute ‘13.1.2.1.1’ for ‘14.1.2.1.1’.

(Page 7, clause **13.1.2.1.2**, line 4) — Substitute ‘IS 15999 Part 1 / IEC 60034-1’ for ‘IS/IEC 60034-1’.

(Page 7, clause **13.2.2**, line 3) — Substitute ‘13.2.2.1’ for ‘14.2.2.1’.

(Page 7, clause **13.2.2.1.1**, line 5) — Substitute ‘IS 15999 Part 1 / IEC 60034-1’ for ‘IS/IEC 60034-1’.

(Page 7, clause **13.2.2.1.2**, line 3) — Substitute ‘13.2.2.1.1’ for ‘14.2.2.1.1’.

(Page 7, clause **13.2.2.1.2**, line 4) — Substitute ‘IS 15999 Part 1 / IEC 60034-1’ for ‘IS/IEC 60034-1’.

(Page 7, clause **14.2**, line 4) — Substitute ‘13’ for ‘14’.

[Page 8, clause **15.2.1**, f) and clause **15.2.2**, d)] — Substitute ‘4m’ for ‘3m’ for the static suction lift of self-priming pumps.

[Page 8, clause **15.2.1**, f) and clause **15.2.2**, d)] — Delete ‘and 1.5 m for semi self-priming pumps’.

[Page 9, clause **15.4.1.1**, e), line 2] — Substitute ‘4m’ in place of ‘3m’.

[Page 9, clause **15.4.1.1**, e), line 3 and 4] — Delete ‘and 1.5 m for semi self-priming pumps’.

(Page 9, clause **15.4.1.1**) — Add the following after e):

‘f) The observed overall efficiency of pumpset/monoset calculated in accordance with following formula –

$$\text{Overall efficiency, } \eta_{ov} = \frac{H \times \dot{Q}}{102 \times \text{Input to motor}}$$

Where,

H = head in metres of water column;

$\dot{Q}$ = discharge in l/s; and

Input to motor in kW.

The overall efficiency as calculated above shall not be less than 0.955 times of the declared value.

NOTE — The overall efficiency declaration and its marking is to create an awareness to the end user regarding the lower efficiency of these type of pumps.’

[Page 11, *clause 15.4.2,c), line 7*] — Substitute ‘**15.4.2 (b)**’ in place of ‘**17.4.2 (b)**’.

[Page 11, *clause 15.4.2, d)*] — Add the following after c):

‘d) The observed pump efficiency of bare pump calculated in accordance to following formula -

$$\text{Pump efficiency, } \eta_p = \frac{H \times \dot{Q}}{102 \times \text{Input to motor}}$$

Where,

H = head in metres of water column;

$\dot{Q}$ = discharge in l/s; and

Input to pump in kW.

The pump efficiency as calculated above shall not be less than 0.95 times of the declared value.

NOTE — The pump efficiency declaration and its marking is to create an awareness to the end user regarding the lower efficiency of these type of pumps.’

(Page 12 *Table 3, Notes 2, line 4*) — Substitute ‘IS 15999 Part 1 / IEC 60034-1’ for ‘IS/IEC 60034-1’.

(Page 13, *Table 4, Notes 3, line 4*) — Substitute ‘IS 15999 Part 1 / IEC 60034-1’ for ‘IS/IEC 60034-1’.

(Page 14, *clause 16.1, t*) — Delete

(Page 14, *clause 16.1, z), line 1*) — Substitute ‘4m’ in place of ‘3m’ for self-priming pump.

(Page 14, *clause 16.1, z), line 2 and 3*) — Delete ‘and 1.5 m for suction lift for semi self-priming pumps’.

(Page 14, *clause 16.1, Notes 1*) — Delete ‘w)’ and renumber the following clauses.

(Page 14, *clause 16.2, l*) — Add ‘Pump Efficiency (percent)’

(Page 15, ANNEX A, *last line*) — Substitute ‘IS 15999 (Part 1) : 2016 / IEC 60034-1: 2010 ‘Rotating electrical machines: Part 1 Rating and performance (*first revision*)’ for ‘IS/IEC 60034-1’ Rotating electrical machines: Part 1 Rating and performance.

At the end of page 15 — Add Page 16 and 17 for ‘ANNEX B’ as mentioned in *Clause 12.6*, ‘Specimen Sheet for IS 8472’.

ANNEX B

(Clause 12.6)

1. SPECIMEN SHEET For IS 8472

MONOSET PUMP TEST RECORD SHEET Sheet No.....											Refer Graph No							
Nature of test – Performance test as per IS 8472																		
Pump type Pump SI No Suction mm Delivery mm Material of impeller											Motor ratingkW/HP Voltage Full load current, Amps .. Meter constant: A ×kW×.....Rated frequency							
Suction lift measured by: Hg manometer/Vacuum gauge Delivery head measured by: Hg manometer/Pressure gauge											Capacity measured by – Vee-notch/Volumetric tank/Flowmeter Class of accuracy of measuring instrument — One							
SI No.	Frequency, Hz	Suction gauge reading, m	Delivery gauge reading, m	Gauge distance, Z, m	Velocity head correction m	Total head, m	Head over notch, mm (in case of Vee-notch)	Volume (l)/time (s) (in case of volumetric tank)	Discharge, l/s	Current A	Wattmeter Reading		Motor input (<i>IP</i>), kW	Pump output (<i>OP</i>), kW	Overall efficiency, percent	Perfor mance converted at rated frequency.....		
											W ₁	W ₂				H <i>m</i>	Q <i>l/s</i>	IP, kW

Pump certified for : i) Total Head in m ii) Discharge in l/s.....Frequency iii) Overall efficiency percentMotor input, kW iv) Head range in m H QGuarantee factor Overall efficiency at duty point, percent General requirements – Satisfactory/Unsatisfactory	DateTested by Remarks Impeller diameter mm
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ANNEX B

(Clause 12.6)

2. SPECIMEN SHEET For IS 8472

BARE PUMP TEST RECORD SHEET											Sheet No.										
Refer Graph No																					
Nature of test – Perform mance test as per IS 8472																					
Pump type Pump SNo Suction mm Delivery mm Material of impeller											Calibrated Test Motor rating kW/HP Voltage Full load current, Amps .. Meter constant: A × kW × Rated frequency										
Suction lift measured by: Hg manometer/Vacuum gauge Delivery head measured by: Hg manometer/Pressure gauge											Capacity measured by – Vee-notch/Volumetric tank/Flowmeter Class of accuracy of measuring instrument — One										
Sl No.	Frequency, Hz	Suction gauge reading, m	Delivery gauge reading, m	Gauge distance, Z, m	Velocity head correction m	Total head, m	Head over notch, mm (in case of Vee-notch)	Volume (l)/time (s) (in case of volumetric tank)	Discharge, l/s	Current A	Wattmeter Reading		Motor input (IP), kW	Motor Efficiency percent	Pump input (IPp), kW	Pump output (OP), kW	Pump efficiency, percent	Perform mance converted at rated frequency.....			
											W ₁	W ₂						H m	Q l/s	IPp, kW	
Pump certified for : i) Total Head in m ii) Discharge in l/s..... Frequency iii) Pump efficiency percent Pump input, kW iv) Head range in m v) Recommended Prime Mover in kW H Q Guarantee factor Pump efficiency at duty point, percent General requirements – Satisfactory/Unsatisfactory											Date Tested by Remarks Impeller diameter mm										